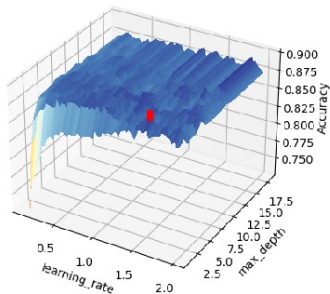
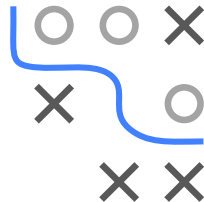


Optimization in Machine Learning

Optimization Problems

Other optimization problems



(a) vehicle

Learning goals

- Discrete / feature selection
- Black-box / hyperparameter optimization
- Noisy
- Multi-objective

OTHER CLASSES OF OPTIMIZATION PROBLEMS

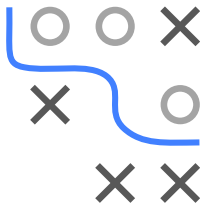
So far: “nice” (un)constrained problems:

- Problem defined on continuous domain $\mathcal{S}$
- Analytical objectives (and constraints)

Other characteristics:

- Discrete domain $\mathcal{S}$
- f **black-box**: Objective not available in analytical form
computer program to evaluate
- f **noisy**: Objective can be queried but evaluations are noisy
 $f(\mathbf{x}) = f_{\text{true}}(\mathbf{x}) + \epsilon, \quad \epsilon \sim F$
- f **expensive**: Single query takes time / resources
- f multi-objective: $f(\mathbf{x}) : \mathcal{S} \rightarrow \mathbb{R}^m, f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))$

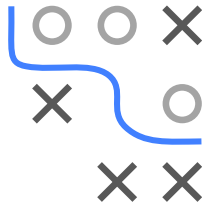
These make the problem typically much harder to solve!



EXAMPLE 1: BEST SUBSET SELECTION

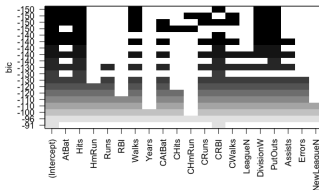
Let $\mathcal{D} = \left(\left(\mathbf{x}^{(i)}, y^{(i)} \right) \right)_{1 \leq i \leq n}$, $\mathbf{x}^{(i)} \in \mathbb{R}^p$. Fit LM based on best feature subset.

$$\min_{\theta \in \Theta} \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)} \right)^2, \|\theta\|_0 \leq k$$



Problem characteristics:

- White-box: Objective available in analytical form
- Discrete: $\mathcal{S}$ is mixed continuous and discrete
- Constrained



► [Click for source](#)

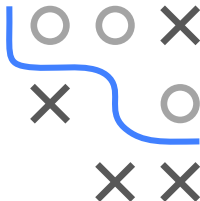
The problem is even **NP-hard** (Bin et al., 1997, The Minimum Feature Subset Selection Problem)!

EXAMPLE 2: WRAPPER FEATURE SELECTION

Subset sel. can be generalized to any learner $\mathcal{I}$ using only features $\mathbf{s}$:

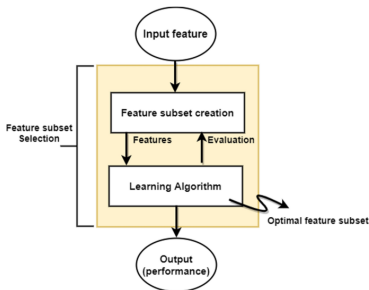
$$\min_{\mathbf{s} \in \{0,1\}^p} \widehat{\text{GE}}(\mathcal{I}, \mathcal{J}, \rho, \mathbf{s}),$$

$\widehat{\text{GE}}$ general. err. with metric ρ and estim. with resampling splits $\mathcal{J}$



Problem characteristics:

- black box
eval by program
- $\mathcal{S}$ is discrete / binary
- expensive
1 eval: 1 or multiple ERM(s)!
- noisy
uses data / resampling
- NB: Less features can be better
in prediction (overfitting)

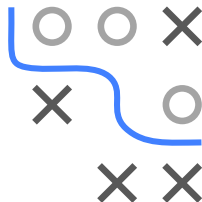


EXAMPLE 3: FEATURE SEL. (MULTIOBJECTIVE)

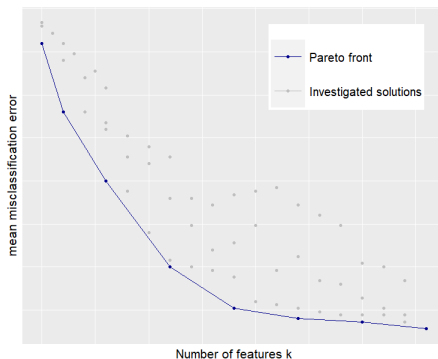
Feature selection is usually inherently multi-objective, with model sparsity as a 2nd trade-off target:

$$\min_{\mathbf{s} \in \{0,1\}^p} \left(\widehat{\text{GE}}(\mathcal{I}, \mathcal{J}, \rho, \mathbf{s}), \sum_{i=1}^p s_i \right).$$

$\widehat{\text{GE}}$ general. err. with metric ρ and estim. with resampling splits $\mathcal{J}$



- Multiobjective
- black box
eval by program
- S is discrete / binary
- expensive
1 eval: 1 or multiple ERM(s)!
- noisy
uses data / resampling

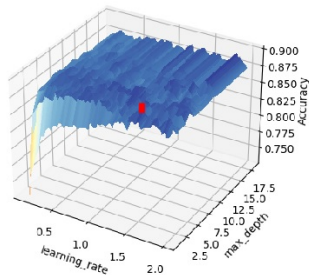
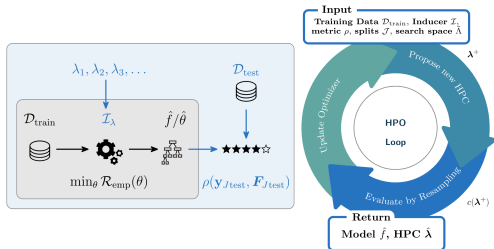
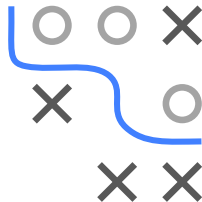


EXAMPLE 4: HYPERPARAMETER OPTIMIZATION

- Learner $\mathcal{I}$ usually configurable by hyperparameters $\lambda \in \Lambda$
- Find best HP configuration λ^*

$$\lambda^* \in \arg \min_{\lambda \in \Lambda} c(\lambda) = \arg \min \widehat{\text{GE}}(\mathcal{I}, \mathcal{J}, \rho, \lambda)$$

$\widehat{\text{GE}}$ general. err. with metric ρ and estim. with resampling splits $\mathcal{J}$



(a) vehicle

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